Network Monopoly: Operator Edition

4 PLAYERS / 8 TURNS EACH / GAME INSTRUCTIONS AND HOST GUIDE

Buy routers. Connect networks. Deliver packets.

Four players act as network operators. Acquire routers, build links and earn resources by delivering packets through the network.

1 Set up the board

Place public routers at P03 and P15 and connect them with a permanent public link. Both routers and this link are free to use. Each player takes one playing piece and 5 resources. All pieces start at P06.

2 Prepare for your turn

Choose the first player randomly, then take turns clockwise. On each turn, draw a movement card marked 1, 2 or 3. Return and shuffle it after use. Move clockwise by that number. Landing on or passing P06 gives no event or bonus.

3 Use the two paths separately

Players may share spaces. Pieces move around the 24-space loop; packets use only the drawn network links. Moving a piece does not create a connection. A corner triggers a task, but the route card determines its source and destination.

How to win

After every player has completed 8 turns, the most remaining resources wins. Routers add no points. Players tied for the highest resource total share the win.

Buy routers and build the network

PURCHASABLE SPACES / P02, P05, P09, P12, P16 AND P21

1 Buy a router and its first link

On an unowned router space, you may pay 3 resources to buy it and connect it to one router already connected to the network. The price includes this first link.

2 Add an alternate route at purchase

At purchase, you may pay 1 extra resource to connect the new router to a second connected router. You may connect to a public router or a router owned by any player.

3 Record links and ownership

Draw player-built links as solid lines in the owner's colour. Crossing lines do not connect unless a router node is at the crossing. Owned routers cannot be bought again; landing on one has no effect. Other players may use connected routers, subject to transit fees.

Fixed access links for the four corner networks

P01 / Network 1	192.168.1.0/24	Gateway: P02
P07 / Network 2	192.168.2.0/24	Gateway: P09
P13 / Network 3	192.168.3.0/24	Gateway: P16
P19 / Network 4	192.168.4.0/24	Gateway: P21

Gateways must be purchased

Each corner network has a fixed access link to its gateway position. It joins the network when that gateway is purchased. If either endpoint gateway is unpurchased, or no valid route exists, the task fails.

Packet tasks and transit fees

DRAW A ROUTE CARD + A RESOURCE CARD / ENDPOINTS AND REWARD

1 Draw two cards

At a packet-task corner, draw one route card and one resource card. Source and destination use different /24 networks. Resource cards show 4, 5 or 6. When a deck runs out, shuffle and reuse its discard pile.

2 Find a valid route

Find a path from the source network's gateway to the destination network's gateway. Every router must be public or purchased and connected. No active ACL may block the route.

3 Receive resources, then pay fees

For a valid route, receive the resource-card amount, then pay transit fees. Each privately owned router on the route charges 1 resource. Public routers are free. The total is capped at 2 resources per packet: pay the first two applicable fees from source to destination.

4 No redraw after failure

With no valid route, the task fails and you receive no resources. Do not redraw when a gateway is unpurchased or a route is blocked. Experiment 1 covers the same-subnet comparison.

P23 / INSTANT PACKET

Draw and resolve cards as for a normal packet task, but the route may pass through no more than three routers.
This limit applies to P23 only. Corner tasks use the valid-route rules above.

Space events at a glance

RESOLVE THE EVENT ON THE SPACE WHERE YOUR PIECE LANDS

Buy router

P02 / P05 / P09 / P12 / P16 / P21

If unowned, you may buy and connect it for 3 resources. A second link costs 1 extra resource at purchase.

Public router

P03 / P15

Available to everyone, with no transit fee.

Question

P04 / P10 / P17 / P22 / P24

Correct: choose +2 resources or move forward 1 space without triggering its event. Incorrect: no reward or penalty.

Router transfer

P18

Pay another player 3 resources to take one of their routers. They must retain at least one router. If no router is eligible or you cannot pay, nothing happens. Existing links remain unchanged.

ACL

P11

Place an ACL on an active link and name a blocked source network. Packets from that source cannot use this link. Maximum one ACL per link. Remove it at the start of your next turn. If no link is eligible, nothing happens.

Instant packet

P23

Use normal packet-task rules, with a maximum of three routers on the route.

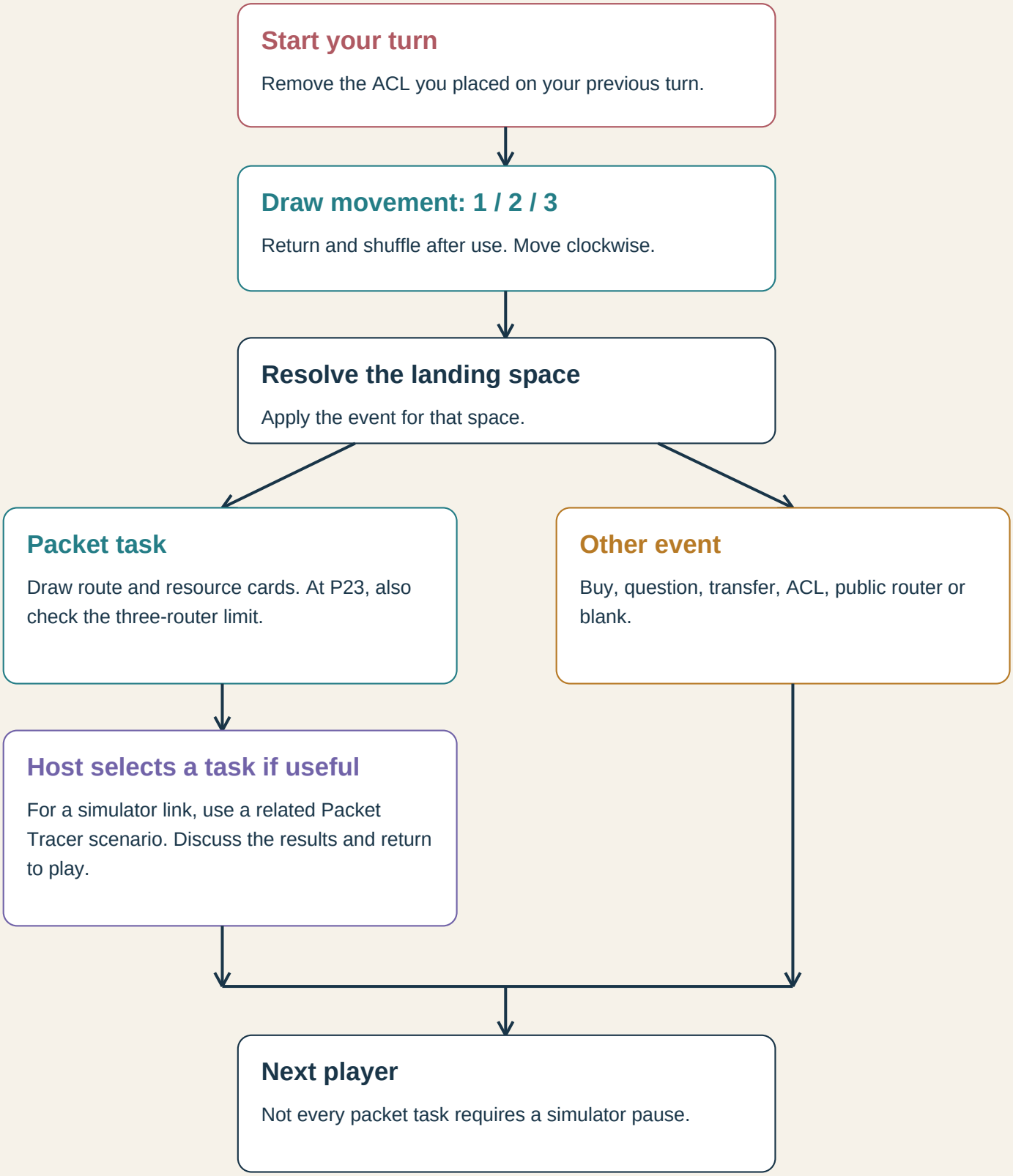
Blank / Start

P06 / P08 / P14 / P20

No effect. Passing or landing on Start gives no bonus.

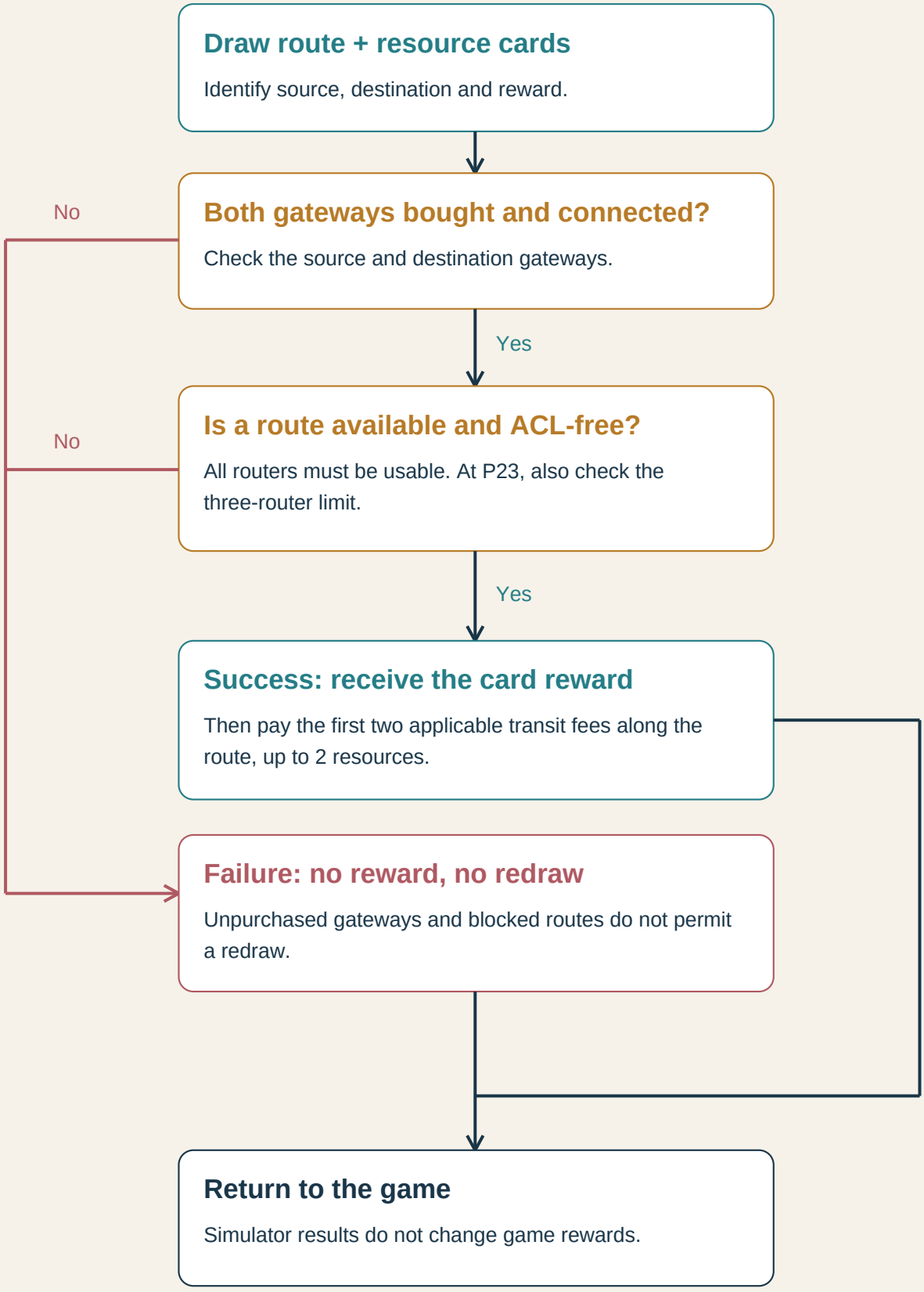
Your turn, step by step

DRAW / MOVE / RESOLVE / PASS TO THE NEXT PLAYER



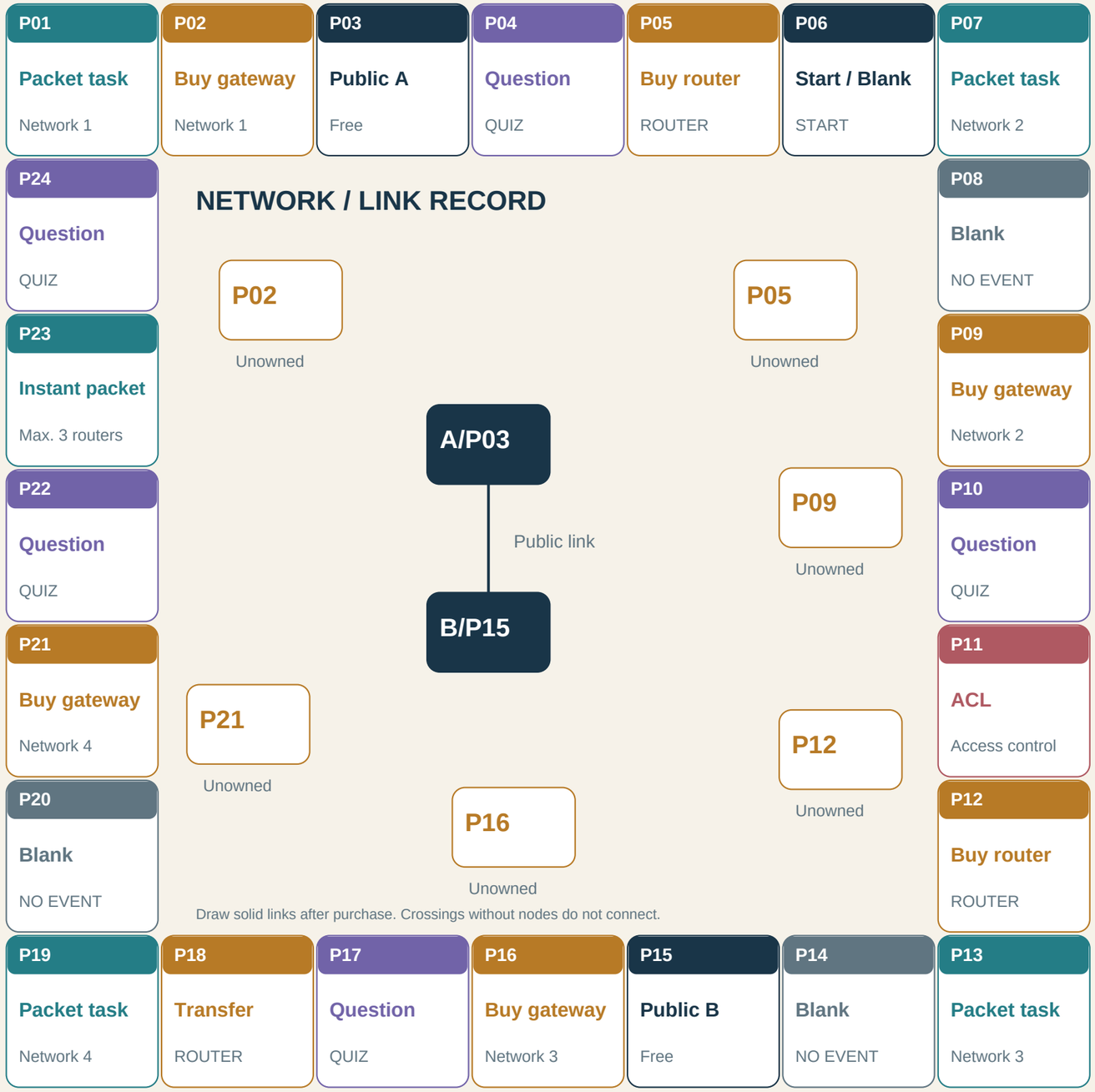
Resolve a packet task

CHECK GATEWAYS / CONNECTED ROUTE / ACL



Network Monopoly / 24-space board

OUTER LOOP: PIECES / CENTRE: NETWORK LINKS / START AT P06 AND MOVE CLOCKWISE



Three steps each turn

01 Draw movement: 1 / 2 / 3

02 Move clockwise

03 Resolve the landing space

Starting network

Only public A, public B and their public link are connected.
All six private routers await purchase. Central nodes record the outer-space devices; they are not extra routers.

Fixed access links

P01 192.168.1.0/24 > Gateway P02

P07 192.168.2.0/24 > Gateway P09

P13 192.168.3.0/24 > Gateway P16

P19 192.168.4.0/24 > Gateway P21

ROUND / Advance after every player has taken one turn

1	2	3	4	5	6	7	8

Router cards

SIX PURCHASABLE LOCATIONS / PURCHASE COST AND LINKS

P02 / Network 1 gateway

Cost: 3 resources, including the first link

Connect to one connected router.
At purchase, pay +1 for a second link.
Others may use it; transit fees apply.

Owner: _____

Record links in the central network area.

P05 / Router

Cost: 3 resources, including the first link

Connect to one connected router.
At purchase, pay +1 for a second link.
Others may use it; transit fees apply.

Owner: _____

Record links in the central network area.

P09 / Network 2 gateway

Cost: 3 resources, including the first link

Connect to one connected router.
At purchase, pay +1 for a second link.
Others may use it; transit fees apply.

Owner: _____

Record links in the central network area.

P12 / Router

Cost: 3 resources, including the first link

Connect to one connected router.
At purchase, pay +1 for a second link.
Others may use it; transit fees apply.

Owner: _____

Record links in the central network area.

P16 / Network 3 gateway

Cost: 3 resources, including the first link

Connect to one connected router.
At purchase, pay +1 for a second link.
Others may use it; transit fees apply.

Owner: _____

Record links in the central network area.

P21 / Network 4 gateway

Cost: 3 resources, including the first link

Connect to one connected router.
At purchase, pay +1 for a second link.
Others may use it; transit fees apply.

Owner: _____

Record links in the central network area.

Route cards / 1

PRINTABLE CARDS / PREPARE DECKS AS NEEDED FOR THE ACTIVITY

Network 1 > Network 2

SOURCE NETWORK

192.168.1.0/24

DESTINATION NETWORK

192.168.2.0/24

Also draw a resource card. The corner does not determine the endpoints.

Network 1 > Network 3

SOURCE NETWORK

192.168.1.0/24

DESTINATION NETWORK

192.168.3.0/24

Also draw a resource card. The corner does not determine the endpoints.

Network 1 > Network 4

SOURCE NETWORK

192.168.1.0/24

DESTINATION NETWORK

192.168.4.0/24

Also draw a resource card. The corner does not determine the endpoints.

Network 2 > Network 1

SOURCE NETWORK

192.168.2.0/24

DESTINATION NETWORK

192.168.1.0/24

Also draw a resource card. The corner does not determine the endpoints.

Network 2 > Network 3

SOURCE NETWORK

192.168.2.0/24

DESTINATION NETWORK

192.168.3.0/24

Also draw a resource card. The corner does not determine the endpoints.

Network 2 > Network 4

SOURCE NETWORK

192.168.2.0/24

DESTINATION NETWORK

192.168.4.0/24

Also draw a resource card. The corner does not determine the endpoints.

Route cards / 2

PRINTABLE CARDS / PREPARE DECKS AS NEEDED FOR THE ACTIVITY

Network 3 > Network 1

SOURCE NETWORK

192.168.3.0/24

DESTINATION NETWORK

192.168.1.0/24

Also draw a resource card. The corner does not determine the endpoints.

Network 3 > Network 2

SOURCE NETWORK

192.168.3.0/24

DESTINATION NETWORK

192.168.2.0/24

Also draw a resource card. The corner does not determine the endpoints.

Network 3 > Network 4

SOURCE NETWORK

192.168.3.0/24

DESTINATION NETWORK

192.168.4.0/24

Also draw a resource card. The corner does not determine the endpoints.

Network 4 > Network 1

SOURCE NETWORK

192.168.4.0/24

DESTINATION NETWORK

192.168.1.0/24

Also draw a resource card. The corner does not determine the endpoints.

Network 4 > Network 2

SOURCE NETWORK

192.168.4.0/24

DESTINATION NETWORK

192.168.2.0/24

Also draw a resource card. The corner does not determine the endpoints.

Network 4 > Network 3

SOURCE NETWORK

192.168.4.0/24

DESTINATION NETWORK

192.168.3.0/24

Also draw a resource card. The corner does not determine the endpoints.

Movement and resource cards

MOVE 1 / 2 / 3 SPACES / REWARD 4 / 5 / 6 RESOURCES / PRINT AS NEEDED

MOVEMENT

1

Move clockwise

Return and shuffle after use.

MOVEMENT

2

Move clockwise

Return and shuffle after use.

MOVEMENT

3

Move clockwise

Return and shuffle after use.

RESOURCES

4

Earn on success

Receive resources, then pay transit fees up to a total of 2.

RESOURCES

5

Earn on success

Receive resources, then pay transit fees up to a total of 2.

RESOURCES

6

Earn on success

Receive resources, then pay transit fees up to a total of 2.

Question cards and answers

FIVE QUESTIONS / KEEP THE HOST ANSWER PANEL OUT OF THE DECK

QUESTION 1

Correct: +2 resources OR move 1 space

What is the subnet mask for /24?

If moving, do not trigger the new event.

QUESTION 2

Correct: +2 resources OR move 1 space

Are 192.168.1.10/24 and 192.168.1.200/24 in the same subnet?

If moving, do not trigger the new event.

QUESTION 3

Correct: +2 resources OR move 1 space

Are 192.168.1.10/24 and 192.168.2.10/24 in the same subnet?

If moving, do not trigger the new event.

QUESTION 4

Correct: +2 resources OR move 1 space

A PC at 192.168.1.10/24 sends to 192.168.1.20. Does it send to its default gateway first?

If moving, do not trigger the new event.

QUESTION 5

Correct: +2 resources OR move 1 space

A PC at 192.168.1.10/24 sends to 192.168.2.20. Which next-hop device does it use?

If moving, do not trigger the new event.

HOST ANSWERS / Not in deck

1 255.255.255.0

2 Yes.

3 No.

4 No. It is in the same subnet.

5 Its default gateway.

Player panels and ACL markers

TRACK RESOURCES / COMPLETED TURNS / ACTIVE ACLS

PLAYER 1

Starting resources: 5

Resources: _____

Completed turns: ____ / 8

Routers do not add points.

PLAYER 2

Starting resources: 5

Resources: _____

Completed turns: ____ / 8

Routers do not add points.

PLAYER 3

Starting resources: 5

Resources: _____

Completed turns: ____ / 8

Routers do not add points.

PLAYER 4

Starting resources: 5

Resources: _____

Completed turns: ____ / 8

Routers do not add points.

ACL / BLOCK SOURCE

Place on an active network link

Set by: ____ Source network: ____
Packets from this source cannot pass.
Remove at the start of the setter's next turn.

ACL / BLOCK SOURCE

Place on an active network link

Set by: ____ Source network: ____
Packets from this source cannot pass.
Remove at the start of the setter's next turn.

Maximum one active ACL per link. No eligible link means no effect. Copy markers as needed.

From the board to Packet Tracer

SELECT REPRESENTATIVE TASKS / SIMULATOR RESULTS DO NOT CHANGE GAME REWARDS

1 Two types of cards

Board-game task cards determine the source network, destination network and packet resources. Simulator scenario cards guide participants through a related network test in Packet Tracer.

2 The host selects a scenario

After players discuss or resolve a representative packet task, introduce a related scenario. Use Experiment 1 for subnet communication and Experiment 2 for routes across multiple routers.

3 Test, compare and return

Participants complete the simulator task, compare the result with the board-game situation, then return to play. Simulator results do not change the game task result or resource award.

Host transition prompt

"We have considered whether this packet can reach its destination on the game board. Now let's test a related network situation in Packet Tracer and see what happens."

Suggested timing

Game: 30-40 minutes. Experiment 1: 10-12 minutes. Experiment 2: 15-20 minutes. Closing discussion: about 5 minutes. Main activity: approximately 65-85 minutes. Optional VLAN and ACL experiments add 10-15 minutes each.

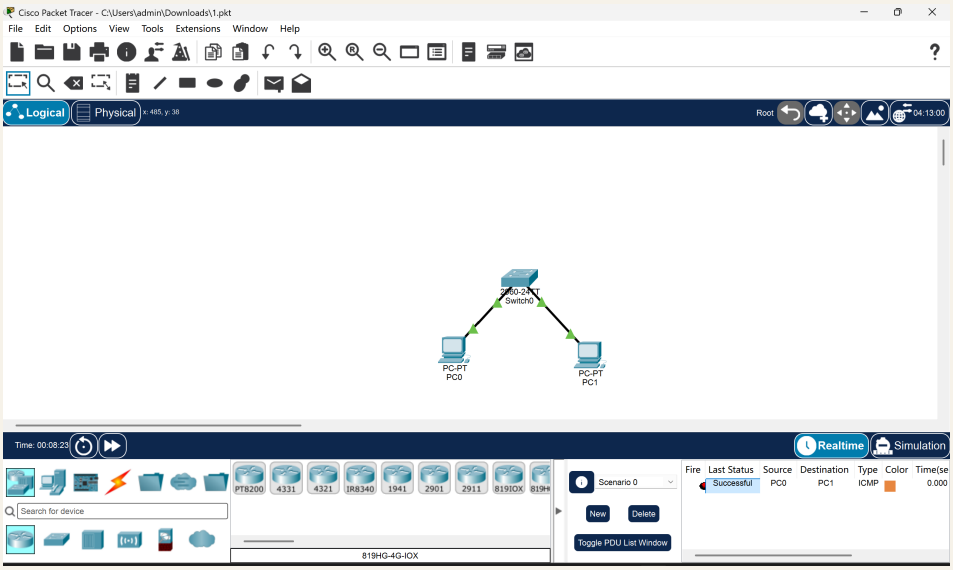
Experiment 1 / Subnet communication

SAME SUBNET AND DIFFERENT SUBNETS / CAN PC0 REACH PC1?

Scenario card

PC0 and PC1 connect to the same switch. Test two addressing scenarios supplied by the host:
A: The PCs are in different subnets.
B: The PCs are in the same /24 subnet.
Send a packet from PC0 to PC1 in each scenario. Record success or failure, compare the addressing and explain the results.

Packet Tracer test view



Host notes

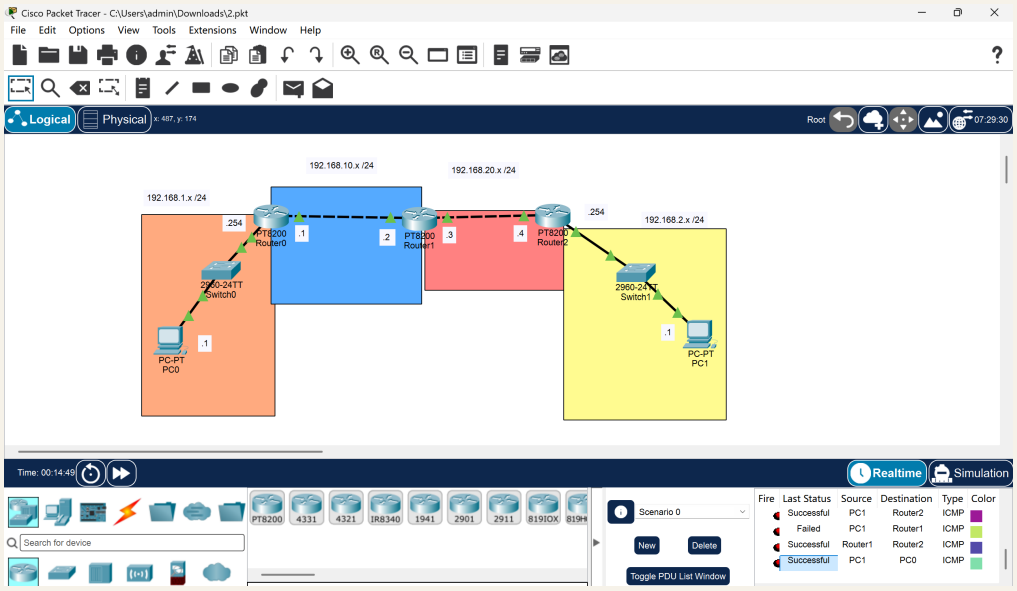
In the 1b.pkt test, communication failed across different subnets. It succeeded after both PCs were changed to the same /24 subnet. The image shows the 1.pkt topology and successful test.

Experiment 2 / Static routing

DELIVER A PACKET THROUGH THREE ROUTERS

Scenario card

PC0 must reach PC1 through Router0, Router1 and Router2.
PC0 LAN: 192.168.1.0/24. R0-R1: 192.168.10.0/24.
R1-R2: 192.168.20.0/24. PC1 LAN: 192.168.2.0/24.
Configure static routes using the topology and device addresses. Make the packet reach its destination and record the end-to-end test result.



Host notes

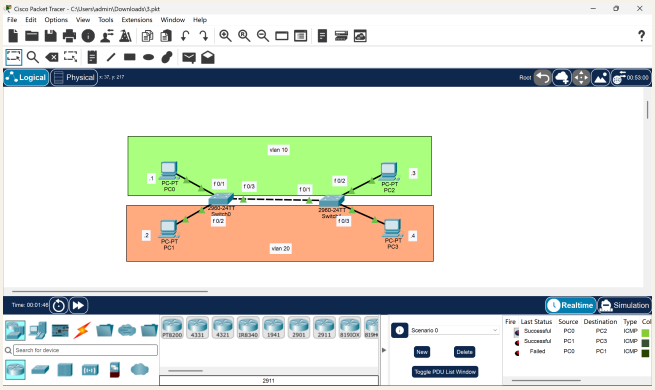
PC0: 192.168.1.1. Router0 LAN: 192.168.1.254.
R0 and R1 use .1 and .2 on 192.168.10.0/24.
R1 and R2 use .3 and .4 on 192.168.20.0/24.
Router2 LAN: 192.168.2.254. PC1: 192.168.2.1.
The 2.pkt test record shows successful end-to-end communication between PC0 and PC1.

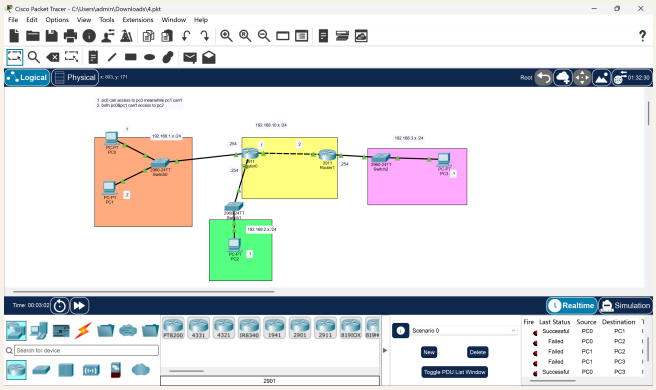
Optional extensions / VLAN and ACL

OPTIONAL EXPERIMENTS / THE P11 BOARD SPACE STILL USES THE ACL GAME RULE

Experiment 3: Which PCs can communicate?

VLAN 10: PC0 and PC2. VLAN 20: PC1 and PC3.
Test PC0 to PC2, PC1 to PC3, and PC0 to PC1. Record which succeed and relate the results to the VLAN assignments.
3.pkt test record: PC0 to PC2 and PC1 to PC3 succeed; PC0 to PC1 fails.





Experiment 4: Which traffic does the ACL allow?

Source LAN: 192.168.1.0/24, with PC0 (192.168.1.1) and PC1 (192.168.1.2).
Destination LAN: 192.168.2.0/24, with PC2 (192.168.2.1).
Destination LAN: 192.168.3.0/24, with PC3 (192.168.3.1).
Test PC0 to PC3, PC1 to PC3, PC0 to PC2, and PC1 to PC2. Record successes and failures and compare them with the topology's access policy.
Policy: PC0 may access PC3; PC1 may not. Neither PC0 nor PC1 may access PC2. The 4.pkt test results match this policy.

Reading and printing guide

RULES / PRINTABLE COMPONENTS / HOST GUIDE

01-04 — **Setup, network and events**

Read these pages before the first game.

05-06 — **Turn and packet flowcharts**

Quick reference for turns and task resolution.

07 — **24-space board**

A3 landscape. Only the A-B public link starts active.

08-13 — **Cards and player panels**

A4. Cut along the frames; print quantities as needed.

14-17 — **Host guide and scenarios**

Simulator activities, test records and discussion.

Printing tips

Print the board on landscape A3 and other pages on A4. Cut cards along the frames. Use player panels to record resources and turns. The host keeps the answer panel and host guide.